

What is a digital transformation?

The term "digital transformation" has become prominent in recent years, but what are its key components? How does it relate to cloud technology, and why do so many organizations pursue it?

The following video answers these questions by examining the history of innovation. Let's explore how the cloud represents a new wave of technology that's reshaping how businesses operate and create value, much like the invention of electricity.

Click play to start the video.



Innovations, paradigm shifts, and digital transformation

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Transcript

Innovation doesn't come in a linear pattern. It comes in waves. And each of these waves is powered by a breakthrough technology. There was the age of the printing press, the steam engine, electricity, the transportation age, the first computers and, today, data and cloud infrastructure. Each of these inventions triggered thousands of innovations, changing what's possible in life and work.

Consider the invention of the printing press. It was revolutionary because it gave everyone access to books, encyclopedias, and even playing cards in their daily lives. It also led to a

broader recognition of intellectual property through widely distributed patents, which in turn prepared the world for the first industrial revolution. There was no turning back!

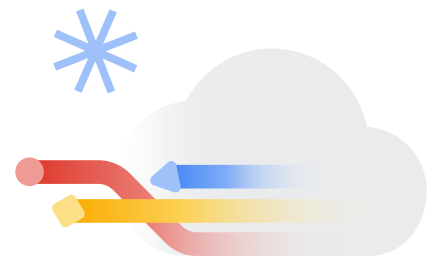
Steam-powered engines brought us cars and trains, which then radically transformed the transportation industry, allowing businesses to produce and transport goods at scale. The entire Industrial Revolution resulted from new technologies that came together and facilitated new ways of working. In the same way, electricity brought us the light bulb, household appliances, and eventually the computer. What the printing press, the steam engine, and electricity all have in common is that they're examples of a paradigm shift: a fundamental and irreversible change in the way that humans work and engage with the world.

How do these examples relate to cloud technology? Well, we're right in the middle of another paradigm shift: one of digital transformation. Cloud technology is transforming how organizations create value, how people work, and ultimately, how people live. It's the catalyst for thousands of innovations that change how we navigate the world, how we interact with media, how we diagnose illness, or how we combat environmental issues.

Defining digital transformation

At Google Cloud, we define digital transformation as:

When an organization uses modern digital technologies—including all types of public, private, and hybrid cloud platforms—to create or modify business processes, culture, and customer experiences to meet changing business and market dynamics.



Organizations choose digital transformation frameworks to **foster innovation, generate new revenue streams, and adapt quickly to market changes and customer needs**. Digital transformation helps organizations change how they operate and redefine relationships with their customers, employees, and partners by modernizing their applications, creating new services, and delivering value. For that reason, rapid advances in digital technology are redefining every industry.

Why it's critical to transform and embrace new technology

As the world and businesses change, keeping technology the same instead of being open to transforming is risky for an organization. Let's illustrate this by looking at two examples: one embraces new technology and uses it to their advantage, and the other doesn't.

Click play to start the video.



Real-world examples: Why it's critical to transform and embrace n

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Nintendo has been creating games since 1889! They started with traditional Japanese playing cards, called Hanafuda, which were made possible by the printing press. From there, they have consistently used new technology to transform their business, and become a leader in the gaming industry. They were even among the first to introduce gaming consoles and mobile gaming devices.

Still, they didn't just stop after these successes. Instead, they revolutionized mobile gaming when they launched Pokemon Go in 2016, and then the first cloud gaming console—Nintendo Switch—one year later in 2017. At a time when most of their competitors were failing, Nintendo transformed by using one new technology after the next, consistently maintaining and even expanding its market share and customer base along the way. More recently, Nintendo has been using Google Cloud to bring games to smartphones worldwide.

So, what makes Nintendo so successful at transforming?

The answer is that they're consistently focused on “why” they exist and not “how” they operate. They exist because they want people to play, and naturally, they'll use any new technology as a resource to achieve this mission. If they focused on liquid crystal displays, or LCDs, as the best tool for gaming, then each new technology would have posed a threat to them. Instead, they used LCDs for a while, and then quickly shifted as the next technology became available to continue motivating people to play.

By contrast, companies that sold encyclopedias, for example, all focused on “how” they operate—how to print and sell a specific set of books. And this was what they were proud of: a beautiful set of leather-bound books lined up on shelves of the finest libraries. And because of their high cost, only a few scholars or elite could afford them. For businesses

that made and sold encyclopedias, they needed printing presses, well-kept warehouses, bookshelf makers, a way to ship and receive heavy containers, and a good door-to-door sales model.

These companies were so focused on manufacturing books that they lost sight of their initial mission: to capture, catalog, and share human knowledge. When new technology such as CD-ROMs became available, this short-term mindset led many of the original encyclopedia companies to view the technology as a threat to their business instead of an opportunity. Ironically, many of the CD-ROM-based encyclopedia providers made the same exact mistake and were later overtaken by cloud-based applications, such as Wikipedia, or traditional encyclopedia companies, such as Britannica, which moved online.

Nintendo and encyclopedia companies were both born from the printing era; Nintendo began with traditional playing cards, and encyclopedias stemmed from hard copy books. Because they reacted to technology innovations differently, they experienced different outcomes.

The reality is that digital transformation is an ongoing process, not a one-time effort. Today, countless industries around the world are disrupted by digitalization: from healthcare to entertainment, from retail to manufacturing.

It's critical that organizations embrace new technology as an opportunity to evolve, serve their customers better, and gain a competitive advantage. This is where cloud computing plays a significant role.